Day 1
Neural network foundations

- neurons and MLPs
- losses, gradients
- backpropagation

Day 2
Training neural networks reliably

- initialization
- regularization
- optimizers, tuning

Day 3
Energy modelling capstone

- data contracts
- tabular-NN playbook
- error analysis

Day 4
Convolutional neural networks

- convolution, pooling
- transfer learning
- 1D CNNs, imaging

Day 5
Sequence models and attention

- windows, baselines
- RNN, LSTM, TCN
- attention, intervals

the evidence chain used every day

frame the question → data contract → split that matches use → baseline first → model and diagnose → report with uncertainty